Fundamentals of Engineering Mathematics I

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Class 9; Series solutions of ODEs 1
Power series method
Convergence of power series
Legendre's equation

We are looking for the solutions of the ODE having variable coefficients.
There are two methods

- Power series method and
- Frobenius method

Power series method

We are looking for a solution in the form of power series.

A power series function expanded at a point x_0 is given or defined by

$$f(x)=\sum_{m=0}^{\infty} a_m (x-x_0)^m = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots$$

with

- non-negative integer m
- constant coefficients a_m
- center of the series x_0

Ex)

$$\frac{1}{1-x}=\sum_{m=0}^{\infty} x^m = 1 + x + x^2 + \cdots$$

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

Further,

$$e^{i\theta}=\sum_{m=0}^{\infty}\frac{(i\theta)^m}{m!}=1+(i\theta)+\frac{(i\theta)^2}{2}+\frac{(i\theta)^3}{6}+\cdots$$

$$=1+\frac{(i\theta)^2}{2!}+\frac{(i\theta)^4}{4!}+(i\theta)+\frac{(i\theta)^3}{3!}+\frac{(i\theta)^5}{5!}+\cdots$$

$$=\left\{1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-\cdots\right\}+i\left\{\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-\cdots\right\}=\cos\theta+i\sin\theta$$

Ex)

For an ODE of

$$y'-y=0 \quad \text{or} \quad y'=y$$

we set a power series function as a trial solution of the ODE as

$$y=\sum_{m=0}^{\infty}a_m x^m=a_0+a_1x+a_2x^2+a_3x^3+\cdots$$

Then, its derivative becomes

$$y'=a_1+2a_2x+3a_3x^2+\cdots$$

Substituting these functions into the ODE gives

$$y'-y=0 \quad\Rightarrow\quad a_1+2a_2x+3a_3x^2+\cdots-a_0-a_1x-a_2x^2-a_3x^3+\cdots=0$$

Collecting the terms of a like power of x gives

$$(a_1-a_0)+(2a_2-a_1)x+(3a_3-a_2)x^2+(4a_4-a_3)x^3+\cdots=0$$

Because the power functions are linearly independent, every coefficient should vanish, giving

$$a_1-a_0=0 \quad\Rightarrow\quad a_1=a_0$$

$$2a_2-a_1=0 \quad\Rightarrow\quad 2a_2=a_1 \quad\Rightarrow\quad a_2=\frac{a_1}{2}=\frac{a_0}{2!}$$

$$3a_3-a_2=0 \quad\Rightarrow\quad 3a_3=a_2 \quad\Rightarrow\quad a_3=\frac{a_2}{3}=\frac{1}{3}\frac{a_0}{2!}=\frac{a_0}{3!}$$

$$4a_4-a_3=0 \quad\Rightarrow\quad 4a_4=a_3 \quad\Rightarrow\quad a_4=\frac{a_3}{4}=\frac{1}{4}\frac{a_0}{3!}=\frac{a_0}{4!}$$

Thus, by reassembling, the solution is obtained simply as

$$y=\sum_{m=0}^{\infty}a_m x^m=a_0+a_0x+\frac{a_0}{2!}x^2+\frac{a_0}{3!}x^3+\frac{a_0}{4!}x^4+\cdots=a_0e^x$$

It is an exponential function!

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Ex) Legendre equation in spherical symmetry

For an ODE of (a Legendre equation in spherical symmetry)

$$\left(1-x^2\right)y''-2xy+2y=0$$

we set the trial solution as

$$y=\sum_{m=0}^{\infty}a_m x^m=a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+a_6x^6+\cdots$$

Then, its derivatives are calculated as

$$y'=a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+6a_6x^5+\cdots$$

$$y''=2a_2+3\cdot 2a_3x+4\cdot 3a_4x^2+5\cdot 4a_5x^3+6\cdot 5a_6x^4+\cdots$$

Substituting these into the ODE, we have

$$y''=2a_2+3\cdot 2a_3x+4\cdot 3a_4x^2+5\cdot 4a_5x^3+6\cdot 5a_6x^4+\cdots$$

$$-x^2y''=-2a_2x^2-3\cdot 2a_3x^3-4\cdot 3a_4x^4-5\cdot 4a_5x^5+\cdots$$

$$-2xy'=-2a_1x-2\cdot 2a_2x^2-2\cdot 3a_3x^3-2\cdot 4a_4x^4-2\cdot 5a_5x^5+\cdots$$

$$2y=2a_0+2a_1x+2a_2x^2+2a_3x^3+2a_4x^4+2a_5x^5+\cdots$$

Collecting like powers gives

$$2a_2+2a_0=0 \quad\Rightarrow\quad a_2=-a_0$$
$$3\cdot 2a_3-2a_2+2a_1=0 \quad\Rightarrow\quad a_3=0$$
$$4\cdot 3a_4-2a_2-2\cdot 2a_2+2a_2=0 \quad\Rightarrow\quad 4\cdot 3a_4-2\cdot 2a_2=0$$
$$\Rightarrow\quad a_4=\frac{1}{3}a_2=-\frac{1}{3}a_0$$
$$5\cdot 4a_5-3\cdot 2a_3-2\cdot 3a_3+2a_3=0 \quad\Rightarrow\quad 5\cdot 4a_5=0 \quad\Rightarrow\quad a_5=0$$
$$6\cdot 5a_6-4\cdot 3a_4-2\cdot 4a_4+2a_4=0 \quad\Rightarrow\quad 30a_6-18a_4=0$$
$$\Rightarrow\quad a_6=\frac{18}{30}a_4=-\frac{18}{30}\frac{1}{3}a_0=-\frac{1}{5}a_0$$

Thus, we have the solution as

$$y=\sum_{m=0}^{\infty}a_m x^m = a_0 + a_1 x - a_0 x^2 + 0x^3 - \frac{1}{3}a_0 x^4 + 0x^5 - \frac{1}{5}a_0 x^6 + \cdots$$
$$= a_1 x + a_0\left\{1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 + \cdots\right\} \equiv a_1 x - a_0 Q_1(x)$$

The solution is given as a linear sum of two elementary functions x and $Q_1(x)$.

It is noted that

- The ODE is the Legendre equation of order 1
- x is a member of Legendre polynomials, $P_1(x)$, and
- $Q_1(x)$ is a member of Legendre functions.

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Convergence of power series

The power series solution of an ODE of

$$y'' + p(x)y' + q(x)y = r(x)$$

converges when all the variable coefficients are **analytic**.

A function f is called analytic when the function can be represented by a power series with positive radius of **convergence**.

If a partial sum s_n of a series s satisfies, at a point x_1, the condition of

$$\lim_{n\to\infty} s_n(x_1) = s(x_1) \quad \text{with} \quad s_n(x) = \sum_{m=0}^{n} a_m x^m$$

then, the series s is called convergent at $x = x_1$.

For an analytic function f, we can approximate the function with a partial sum s_n as accurately as we want with increasing n.

The convergence interval from the center is called the **radius of convergence**.
It is determined by the ratio of nearby coefficients of the series as

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \quad \text{or} \quad R = 1/\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$$

Ex)
For a power series

$$\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m = 1 + x + x^2 + \cdots$$

we have

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1 \quad \Rightarrow \quad R = 1/1 = 1$$

It converges at $|x| < 1$.

Ex)
For

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

we have

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{n!}{(n+1)!}\right| = \lim_{n\to\infty}\left|\frac{1}{n+1}\right| = 0 \quad \Rightarrow \quad R = 1/0 = \infty$$

It converges at any x.

Ex)
For

$$f = \sum_{m=0}^{\infty} m!\,x^m = 1 + x + 2x^2 + 6x^3 + \cdots$$

we have

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)!}{n!}\right| = \lim_{n\to\infty}|n+1| = \infty \quad \Rightarrow \quad R = 1/\infty = 0$$

It converges only at a point, $x = 0$.

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Legendre's equation

The Legendre's equation arises in the Laplace equation in the spherical coordinates, having a form of

$$\left(1-x^2\right)y''-2xy'+n\left(n+1\right)y=0$$

Or, in a standard form of linear ODE, it is

$$y''-\frac{2x}{1-x^2}y'+\frac{n\left(n+1\right)}{1-x^2}y=0$$

Since both the variable coefficients are analytic at x=0 within a radius 1, we can set the trial function with a power series as

$$y=\sum_{m=0}^{\infty}a_m x^m \quad \text{at} \quad \left|x\right|<1.$$

Substituting the trial power series function into the ODE gives

$$\left(1-x^2\right)\sum_{m=2}^{\infty}m\left(m-1\right)a_m x^{m-2}-2x\sum_{m=1}^{\infty}ma_m x^{m-1}+n\left(n+1\right)\sum_{m=0}^{\infty}a_m x^m=0$$

Collecting the terms of a like power from each summation gives

$$\sum_{m=2}^{\infty}m\left(m-1\right)a_m x^{m-2}-\sum_{m=2}^{\infty}m\left(m-1\right)a_m x^m-2\sum_{m=1}^{\infty}ma_m x^m+n\left(n+1\right)\sum_{m=0}^{\infty}a_m x^m=0$$

We can see that only the 1st term has a different power. The other three terms have the same power. So, we need to express the equation with the same power.

Since the integer m is just a summing parameter, we can adjust it as

$$m-2=s \quad \text{or} \quad m=s+2$$

This adjustment makes the 1st term become

$$\sum_{m=2}^{\infty}m\left(m-1\right)a_m x^{m-2} \quad \Rightarrow \quad \sum_{s=0}^{\infty}\left(s+2\right)\left(s+1\right)a_{s+2}x^s$$

The other three terms can be adjusted to have the same summing parameter s, giving

$$\sum_{s=0}^{\infty}\left(s+2\right)\left(s+1\right)a_{s+2}x^s-\sum_{s=2}^{\infty}s\left(s-1\right)a_s x^s-2\sum_{s=1}^{\infty}sa_s x^s+n\left(n+1\right)\sum_{s=0}^{\infty}a_s x^s=0$$

Collecting like powers and making their coefficients vanish, we have

For $s=0$

$$(0+2)(0+1)a_{0+2}+n(n+1)a_0=0 \quad \Rightarrow \quad 2a_2=-n(n+1)a_0$$

$$\Rightarrow \quad a_2=-\frac{n(n+1)}{2}a_0=-\frac{n(n+1)}{2!}a_0$$

For $s=1$

$$(1+2)(1+1)a_{1+2}-2a_1+n(n+1)a_1=0 \quad \Rightarrow \quad 6a_3=-\{n(n+1)-2\}a_1$$

$$\Rightarrow \quad a_3=-\frac{(n-1)(n+2)}{6}a_1=-\frac{(n-1)(n+2)}{3!}a_1$$

For $s\geq 2$

$$(s+2)(s+1)a_{s+2}-s(s-1)a_s-2sa_s+n(n+1)a_s=0$$

$$\Rightarrow \quad a_{s+2}=-\frac{-s(s-1)-2s+n(n+1)}{(s+2)(s+1)}a_s=-\frac{(n-s)(n+s+1)}{(s+2)(s+1)}a_s$$

It means that any coefficients can be obtained from the first two coefficients a_0 and a_1.

For an example

$$a_4=a_{2+2}=-\frac{(n-2)(n+2+1)}{(2+2)(2+1)}a_2=-\frac{(n-2)(n+3)}{4\cdot 3}\left\{-\frac{n(n+1)}{2!}\right\}a_0$$

$$=+\frac{(n-2)n(n+1)(n+3)}{4!}a_0$$

and

$$a_5=a_{3+2}=-\frac{(n-3)(n+3+1)}{(3+2)(3+1)}a_3=-\frac{(n-3)(n+4)}{5\cdot 4}\left\{-\frac{(n-1)(n+2)}{3!}\right\}a_1$$

$$=-\frac{(n-3)(n-1)(n+2)(n+4)}{5!}a_1$$

Similarly, we have

$$a_6 = a_{4+2} = -\frac{(n-4)(n+4+1)}{(4+2)(4+1)}a_4 = -\frac{(n-4)(n+5)}{6\cdot 5}a_4$$
$$= -\frac{(n-4)(n+5)}{6\cdot 5}\frac{(n-2)n(n+1)(n+3)}{4!}a_0$$
$$= -\frac{(n-4)(n-2)n(n+1)(n+3)(n+5)}{6!}a_0$$

and

$$a_7 = +\frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!}a_1$$

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It gives us two solutions. One is proportional to a_0 and the other proportional to a_1,

$$y = a_0 y_1(x) + a_1 y_2(x).$$

$$y_1(x) = \frac{a_0}{a_0}x^0 + \frac{a_2}{a_0}x^2 + \frac{a_4}{a_0}x^4 + \cdots$$
$$= 1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 - + \cdots$$
$$y_2(x) = \frac{a_1}{a_1}x^1 + \frac{a_3}{a_1}x^2 + \frac{a_5}{a_1}x^4 + \cdots$$
$$= x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - + \cdots$$

When the series is made infinitely, it is valid only at $-1 < x < 1$.
However, when the series is terminated at some order, it becomes valid for all *x*.
The termination happens when *n* is an integer.

From the recurrence relation of

$$a_{s+2} = -\frac{(n-s)(n+s+1)}{(s+2)(s+1)}a_s$$

we can see that when *n* is a nonnegative integer, the coefficient a_{n+2} vanishes, since

$$a_{n+2} = -\frac{(n-n)(n+n+1)}{(n+2)(n+1)}a_n = 0$$

Thus, we have

$$a_{n+2} = a_{n+4} = a_{n+6} \cdots = 0$$

The highest order of the power series becomes the order *n*. It gives the Legendre polynomial of order *n*. It is just one elementary solution of the Legendre's equation.

By assigning the coefficient a_0 or a_1 to have the highest order coefficient a_n as

$$a_n = \frac{(2n)!}{2^n (n!)^2}$$

we have the Legendre polynomial of order *n* as

$$P_n(x) = \sum_{m=0}^{M} (-1)^m \frac{(2n-2m)!}{2^n m!(n-m)!(n-2m)!} x^{n-2m}$$

Where, the integer *M* is defined

$$M = \frac{n}{2} \quad \text{for even } n$$

$$M = \frac{n-1}{2} \quad \text{for odd } n$$

We can check that, for every *n*, it is normalized as

$$P_n(x=1) = 1.$$

It is noted that the Legendre polynomials are orthogonal to each other in the interval of $-1 \le x \le 1$.

In $y = a_0 y_1(x) + a_1 y_2(x)$, we can prove that one is a terminated polynomial but the other is not.

In detail, for even *n*, $y_1(x)$ becomes terminated to have the polynomial of order *n*. Whilst, $y_2(x)$ is a polynomial of infinite order.

However, for odd n, $y_2(x)$ becomes terminated to have the polynomial of order n. Whilst, $y_1(x)$ becomes a polynomial of infinite order instead.

The terminated polynomials are the **Legendre polynomials**.
They are orthogonal to each other!
The others are called **Legendre functions**.